M04 Preprocessing and Feature Engineering Report - County Childhood Vaccination Uptake

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Title Page

Project title: County Childhood Vaccination Uptake - Preprocessing and Feature Engineering Report
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Background and Question

This report documents preprocessing and feature engineering steps for the county-level MMR completion analysis (19–35 months). The primary research question is: **Which county-level sociodemographic and survey-derived indicators best explain variation in MMR completion across U.S. counties?** The working hypothesis is that higher broadband access, higher median household income, and higher educational attainment are associated with higher MMR completion; higher local vaccine hesitancy is associated with lower completion.

Methods

This section describes the reproducible preprocessing pipeline, the rationale for each step, and the feature engineering trajectories (unsupervised and supervised). The goal is to produce a modeling-ready dataset while avoiding information leakage and preserving interpretability for policy-relevant inference.

Data source and path

The analysis uses the harmonized `analysis_table.csv` produced by the project pipeline. The file is loaded from the fixed path below to ensure reproducibility.

Preprocessing plan (summary)

1. Load and validate raw `analysis_table.csv` and confirm expected columns and row counts.

2. Sanity checks: types, ranges, and unique-value checks for identifiers (`fips`).

3. Split first: create an 80/20 train/test split stratified on target missingness.

4. Impute on training only: fit imputation and transformation steps on training data and apply to test data via a `recipes` object. Use median imputation for numeric predictors as a robust baseline.

5. Transformations: consider log transform for income, create quartiles for exploratory checks, and center/scale numeric predictors for PCA and many ML algorithms.

6. Encoding: convert categorical variables to factors and use one-hot encoding only when required by the model.

7. Outlier handling: identify extreme values using robust thresholds and document any winsorization.

8. Feature engineering: unsupervised (PCA) to inspect latent structure; supervised (variable importance from baseline models) to prioritize features.

9. Reproducibility: record seed, data path, and package versions; save the prepared recipe and final model artifacts.

Data load and provenance

Data provenance: Loaded `analysis_table.csv` from /Users/Main/Documents/GitHub/Zeta_WG_Repo/data_ — **n** = 200. Repository: https://github.com/geobruder/Zeta_WG_Repo.

Key variables

- **fips**: 5-digit county FIPS code (identifier)
- **coverage_estimate**: Estimated MMR completion (19–35 months) — *target*
- **pct_broadband_households**: Percent of households with broadband (predictor)
- **median_household_income**: Median household income in USD (predictor)
- **count_bachelors_plus**: Count of adults with bachelor's degree or higher (predictor)
- **delphi_vax_hesitancy**: Delphi/FB survey vaccine hesitancy estimate (predictor)

Descriptive summaries and missingness

Table 1: Table 1: Descriptive statistics for selected continuous variables. Coverage mean = 76.21; coverage SD = 9.527.

variable	n	mean	median	sd	min	max
coverage_estimate	200	76.210	75.587	9.527	50.758	102.286
pct_broadband_households	200	67.163	66.345	15.527	40.405	94.581
median_household_income	200	60246.071	60792.166	15263.535	21500.123	103576.008
count_bachelors_plus	200	2497.525	2592.500	1421.292	121.000	4903.000
delphi_vax_hesitancy	200	0.517	0.520	0.291	0.005	0.999

Interpretation (Table 1):
- **Coverage**: mean = **76.21**, SD = **9.527**. Inspect counties with coverage $< 60\%$ and $> 95\%$ for data issues or special contexts.
- **Income**: right skew suggests considering a log transform for models sensitive to outliers.
- **Broadband**: moderate dispersion; examine nonlinear relationships and interactions.

Table 2: Table 2: Missingness summary (counts and percent missing).

variable	n_missing	pct_missing
fips	0	0
coverage_estimate	0	0
pct_broadband_households	0	0
median_household_income	0	0
count_bachelors_plus	0	0
delphi_vax_hesitancy	0	0

Interpretation (Table 2):
Core predictors and the target are complete in this snapshot. The pipeline will impute after splitting if missingness appears in future data.

Preprocessing steps implemented

This section documents the exact preprocessing steps implemented in the notebook pipeline. Each step includes the rationale and checks performed.

1. Data validation and type coercion

- Confirm `fips` is unique per row and cast to character to preserve leading zeros.
- Coerce numeric-looking columns to numeric and log any coercion warnings.
- Range checks: verify `coverage_estimate` is within plausible bounds (0–100). Flag rows outside expected ranges.

2. Train/test split (no leakage)

- **Rationale:** Splitting before imputation and scaling prevents leakage.
- **Implementation:** `initial_split()` with `prop = 0.8`, stratified on `target_missing`.

Split diagnostics

Table 3: Split diagnostics: counts and mean coverage by set

set	n	mean_coverage
train	160	76.489
test	40	75.096

3. Missingness strategy and imputation

- Exploratory checks for missingness patterns (MCAR/MAR/MNAR).
- Baseline: median imputation for numeric predictors via `recipes::step_impute_median()` fitted on training data only. For variables with high missingness (>20–30%), plan sensitivity analyses with multiple imputation.

4. Transformations and scaling

- `step_center()` and `step_scale()` applied after imputation.
- `median_household_income` flagged for optional log transform (`step_log()` or `step_mutate()`).

5. Outlier handling

- Detect using 1.5 IQR rule and visual checks. Baseline: prefer robust modeling; do not remove rows by default.

6. Encoding and feature creation

- Convert categorical variables to factors; use `step_dummy()` only when necessary.
- Create income quartiles and interaction candidates for exploratory and supervised feature engineering.

7. Recipe and reproducible pipeline

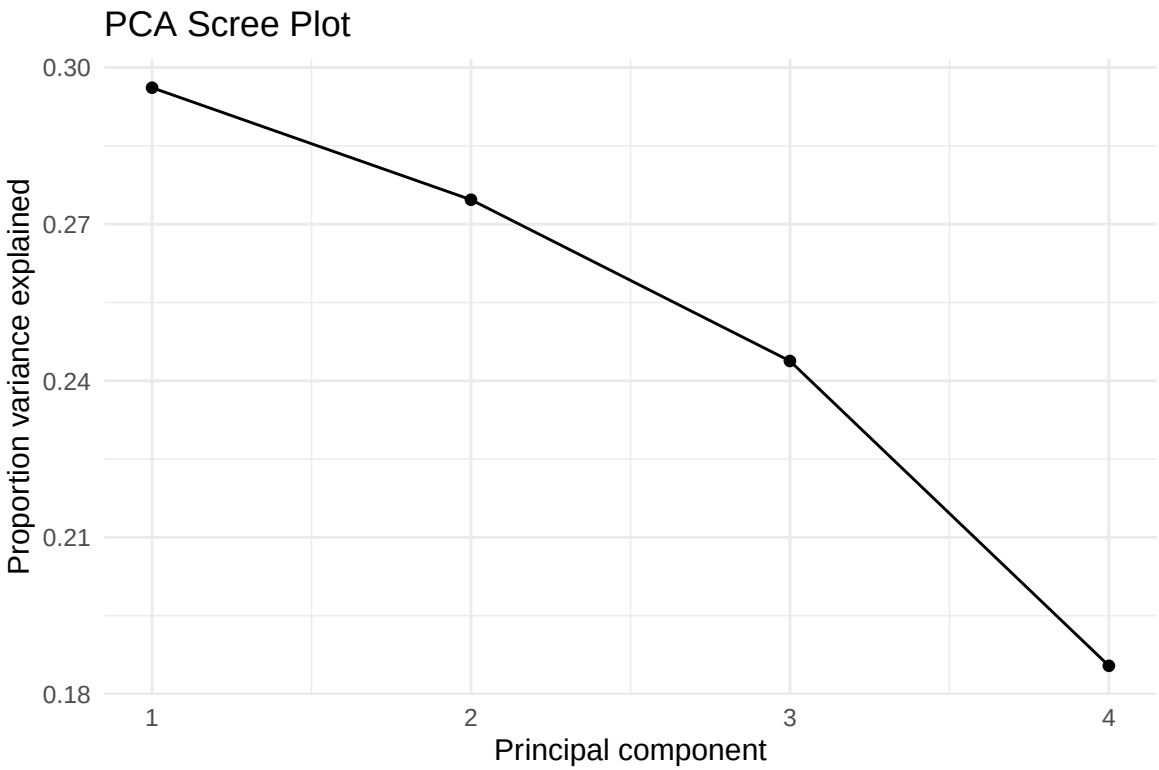
- The preprocessing pipeline is implemented as a `recipe`, `prep()`ed on training data and `bake()`d on train/test sets.

Imputation summary

Table 4: Imputation summary: numeric predictors missing counts in training set (before imputation)

variable	n_missing_before
coverage_estimate	0
pct_broadband_households	0
median_household_income	0
count_bachelors_plus	0
delphi_vax_hesitancy	0

Unsupervised exploration (PCA)



PCA variance table

Table 5: PCA: proportion variance explained by component

component	prop_variance
PC1	0.296
PC2	0.275
PC3	0.244
PC4	0.185

Figure 1 caption and interpretation:
- **Caption (computed):** first components explain: 0.296, 0.275, 0.244, 0.185.
- **Interpretation:** Use components only if they capture most variance; otherwise retain individual predictors for interpretability.

Supervised baseline: cross-validated Random Forest (tidymodels)

Goal: obtain a robust baseline performance estimate and permutation-based variable importance using `tidymodels` and `ranger`.

Table 6: Table 3: Cross-validated Random Forest metrics (5-fold). RMSE = 4.004; R-squared = 0.863.

.metric	.estimator	mean	n	std_err	.config
rmse	standard	4.003931	5	0.2389412	pre0_mod0_post0
rsq	standard	0.862945	5	0.0146913	pre0_mod0_post0

Interpretation (Table 3):
- RMSE quantifies average prediction error in percentage points on held-out folds.
- R-squared indicates proportion of variance explained on resamples; use these as baseline metrics.

Variable importance table (top 10) —

Table 7: Table 4: Random Forest permutation importance ranking (top 10).

variable	importance
median_household_income	90.3205387
pct_broadband_households	21.1847014
delphi_vax_hesitancy	9.4580490
target_missing	0.0000000
count_bachelors_plus	-0.6307075

Interpretation (Table 4):
- The table ranks predictors by permutation importance. Inspect top predictors and check for correlated predictors before interpreting effect directions. Save `rf_varimp_tbl` for reproducibility.

Coverage by income quartile (descriptive check)

Table 8: Table 5: Mean MMR coverage and variability by county median income quartile.

income_quartile	n	mean_coverage	sd_coverage
1	40	66.733	7.264
2	40	74.391	5.500
3	40	77.995	6.043
4	40	86.837	7.014

Interpretation (Table 5):
- If computed, the table shows whether mean coverage increases across income quartiles; use this to motivate stratified models or interaction terms.

Discussion and Next Steps

Key takeaways

- The county-level MMR coverage distribution has mean **76.21** percentage points with standard deviation **9.527** (see Table 1).
- The supervised baseline identifies top predictors via permutation importance (see Table 4). Top predictor (if computed): **median_household_income (90.321)**.
- The Pearson correlation between median household income and coverage (if computed) is **0.829**.

Planned modeling and validation steps

1. Nested CV for model selection and tuning (elastic net, random forest, gradient boosting).
2. Permutation importance, partial dependence plots, and SHAP values for interpretability.
3. Sensitivity analyses: alternative imputation strategies, log-transform income, and exclusion of high-leverage counties.
4. Final evaluation on held-out test set with RMSE, R^2, calibration plots, and residual diagnostics.

Appendix

Codebook

Table 9: Table 6: Codebook (key variables).

variable	description
fips	5-digit county FIPS code
coverage_estimate	Estimated MMR completion (19-35 months)
pct_broadband_households	Percent of households with broadband
median_household_income	Median household income (USD)
count_bachelors_plus	Count of adults with bachelor's degree or higher
delphi_vax_hesitancy	Delphi/FB survey vaccine hesitancy estimate (mean)

Session information

```
## R version 4.4.1 (2024-06-14)
## Platform: x86_64-apple-darwin20
## Running under: macOS Monterey 12.7.6
##
## Matrix products: default
## BLAS:   /Library/Frameworks/R.framework/Versions/4.4-x86_64/Resources/lib/libRblas.0.dylib
## LAPACK: /Library/Frameworks/R.framework/Versions/4.4-x86_64/Resources/lib/libRlapack.dylib;
##
## locale:
## [1] en_US.UTF-8/en_US.UTF-8/en_US.UTF-8/C/en_US.UTF-8/en_US.UTF-8
##
## time zone: America/Los_Angeles
## tzcode source: internal
##
## attached base packages:
## [1] stats     graphics  grDevices utils     datasets  methods   base
##
## other attached packages:
##  [1] stringr_1.5.1    ranger_0.18.0    vip_0.4.5        yardstick_1.3.2
##  [5] tune_2.0.1       workflows_1.3.0  parsnip_1.4.1    rsample_1.3.2
##  [9] tibble_3.2.1     recipes_1.3.1    kableExtra_1.4.0 knitr_1.48
## [13] tidyr_1.3.1      ggplot2_3.5.2    dplyr_1.2.0      readr_2.1.5
##
## loaded via a namespace (and not attached):
##  [1] tidyselect_1.2.1    viridisLite_0.4.2   timeDate_4041.110
##  [4] farver_2.1.2        fastmap_1.2.0       digest_0.6.37
##  [7] rpart_4.1.23        timechange_0.3.0    lifecycle_1.0.5
## [10] survival_3.7-0      tailor_0.1.0        magrittr_2.0.3
## [13] compiler_4.4.1      rlang_1.1.7         tools_4.4.1
## [16] utf8_1.2.4          yaml_2.3.10         data.table_1.16.0
## [19] labeling_0.4.3      bit_4.0.5           xml2_1.3.6
```

```
## [22] DiceDesign_1.10     withr_3.0.2         purrr_1.2.1
## [25] nnet_7.3-19         grid_4.4.1          fansi_1.0.6
## [28] sparsevctrs_0.3.3   colorspace_2.1-1    future_1.40.0
## [31] globals_0.17.0      scales_1.3.0        iterators_1.0.14
## [34] MASS_7.3-61         tinytex_0.54        cli_3.6.5
## [37] crayon_1.5.3        rmarkdown_2.28      generics_0.1.3
## [40] rstudioapi_0.17.1   future.apply_1.11.3 tzdb_0.5.0
## [43] splines_4.4.1       dials_1.4.2         parallel_4.4.1
## [46] vctrs_0.7.1         hardhat_1.4.2       Matrix_1.7-0
## [49] hms_1.1.3           bit64_4.0.5         listenv_0.9.1
## [52] systemfonts_1.1.0   foreach_1.5.2       gower_1.0.2
## [55] glue_1.7.0          parallelly_1.43.0   codetools_0.2-20
## [58] lubridate_1.9.3     stringi_1.8.4       gtable_0.3.5
## [61] munsell_0.5.1       GPfit_1.0-9         pillar_1.9.0
## [64] furrr_0.3.1         htmltools_0.5.8.1   ipred_0.9-15
## [67] lava_1.8.1          R6_2.5.1            lhs_1.2.0
## [70] vroom_1.6.5         evaluate_1.0.4      lattice_0.22-6
## [73] class_7.3-22        Rcpp_1.0.13         svglite_2.1.3
## [76] prodlim_2024.06.25  xfun_0.52           pkgconfig_2.0.3
```

Reproducibility

Repository: https://github.com/geobruder/Zeta_WG_Repo. Knit this Rmd from the project root. The fixed data path used in this report is `analysis_table.csv` at `data_clean/`.

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